

Revised Module-Wise Project Plan

1. Authentication Module

Folder

modules/auth/

Responsibility

Authentication, authorization, and access control.

Features Covered

✓ Authentication & Role-Based Access

2. Academic Module (Core Module)

Folder

modules/academic/

Responsibility

Everything related to academic operations and student records.

Features Covered

- ✓ Student Profile Management
 - ✓ Course Enrollment Management
 - ✓ Attendance Management & Analytics
 - ✓ Grade & Assessment Management
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3. Analytics Module

Folder

modules/analytics/

Responsibility

All dashboard analytics and performance computation.

Features Covered

- ✓ GPA & Academic Performance Analytics
 - ✓ Instructor Analytics Dashboard
 - ✓ Semester Comparison Analytics
 - ✓ Admin Department Analytics
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4. Reporting Module

Folder

modules/reporting/

Responsibility

Generate academic reports.

Features Covered

- ✓ Academic Reports Generation
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5. Intelligence Module

Folder

modules/intelligence/

Responsibility

ML and recommendation features.

Features Covered

- ✓ ML-Based Performance Prediction
 - ✓ Personalized Recommendation Engine
-

Final Module Architecture

modules/



Recommended Development Order

Build in this order:

Phase 1 — Foundation

1. `auth`
2. `academic`

Phase 2 — Dashboards

3. `analytics`

Phase 3 — Reports

4. `reporting`

Phase 4 — AI Features

5. `intelligence`

This is cleaner, more maintainable, and much better aligned with your project size goal.